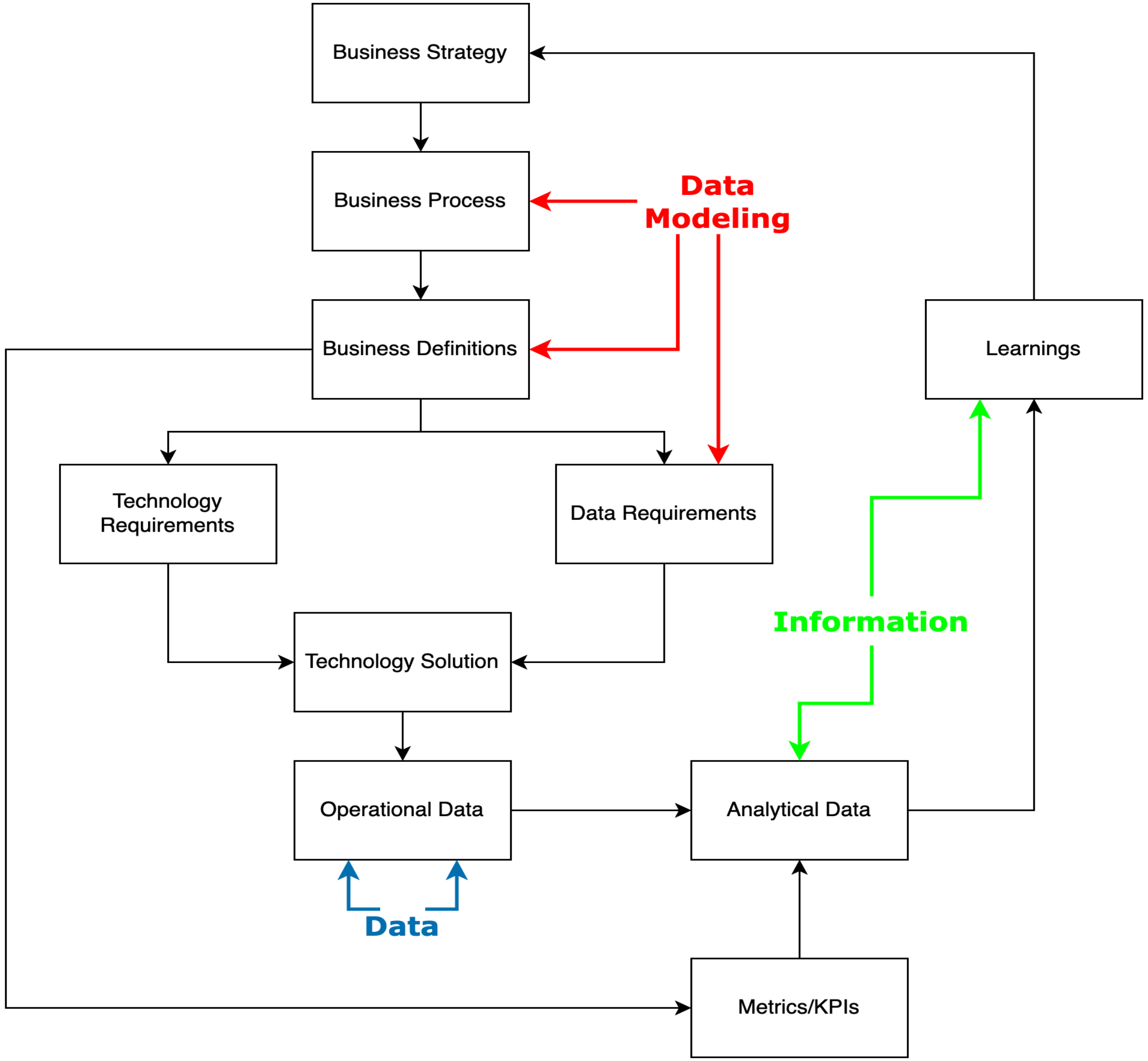
Lesson 13: Business Intelligence and Data Warehousing

What is a Business Intelligence

- The tools and process that turn data into information to:
  - Generate insights
  - Inform process improvements
  - Drive high-level business/operational decisions
  - Make all of the investment into data worthwhile!

The Data Feedback Loop



BI Capabilities

- ETL, ELT and Reverse ETL Tooling
  - Get our Data into a place to transform it into Information
  - Sometimes get it back to an operational system
- Data Storage (Data Warehouse/Mart)
  - Platforms optimized for data synthesis, aggregation and analytics
  - Querying is a must
- Data Viz Tools
  - Think building charts and dashboards
  - Interactivity is nice, but not required

BI Capabilities (Continued)

- Data Monitoring and Alerting
  - Good businesses are proactive rather than reactive
  - Relies on strong processes and definitions on the business teams
- Data Analytics
  - The basics of "how many customers per time period"
  - To the advanced like "based on the history of customers similar to customer X, what do we predict their purchase history to be?"

Data Governance and Management

- Data Governance
    - Establishes the policies and procedures that control data
- Data Management
    - Enacts the policies and procedures so that data can be used for decision making

KPIs

- Key Performance Indicators (KPIs) are numeric measures used for quantifying and measuring against goals
  - This is a must for a data-driven organization

BI Matters

- No matter the sector that you work in, public or private, BI matters
- It is a discipline that is interoperable even if the context itself is different

Types of Data

- **Operational Data** is highly normalized relational data stored by a database optimized for transactions (Data)
- **Decision Support Data** likely denormalized data that is aggregated and used to inform decision making, stored by a data optimized for aggregation and query processing (Information)

Comparing Operational and Decision Support Data (High-Level)

Characteristic	Operational	Decision Support
Time span	Short	Long
Granularity	High	High to Low
Dimensionality	Low	High

Comparing Operational and Decision Support Data (Detailed)

Characteristic	Operational	Decision Support
Currency	Current state, real-time	Prior state snapshot, historic
Volatility	High	Low
DML Focus	Insert/Update	Select
Data Model	Highly normalized, mostly relational	Non-normalized, some relational
Query Activity	Low to Medium	High
Query Complexity	Low	High

Decision Support Database Requirements

- Offers support for complex, de-normalized schemas
- Supports advanced data extraction (large volumes of data) and filtering (think `WHERE`)
- Enables the storage of big data
    - At least data that's probably bigger than the operational data

Data Warehouses and Marts

- A **data warehouse** is an integrated, subject-oriented, time-variant, nonvolatile collection of data that provides support for decision making
- A **data mart** is a smaller, single-subject data warehouse catered to a small group of end users

Twelve Rules of a Data Warehouse (1-6)

The data warehouse...

1. is separate from the operational environment
2. data is integrated
3. contains historic data over a long time period
4. data is a snapshot from a given period of time
5. is subject oriented

Twelve Rules of a Data Warehouse (7-12)

The data warehouse...

7. development life-cycle is data driven based on an organizations data needs
8. contains data with several levels of detail
9. is characterized by read-only transactions to large datasets
10. environment has a system that traces data sources, transformation and storage
11. relies on metadata to identify and define all data elements
12. relies on a charge back mechanism to enforce/encourage optimal usage by end users

Star Schemas

- A **star schema** is a data modeling technique used to map multidimensional decisions support data into a relational database
  - Consists of facts, dimensions and attributes

Star Schema Components

- **Facts** are numeric measurements that represent a specific business or operational activity
- **Dimensions** are qualifying characteristics that provide additional perspectives to a given fact
- **Attributes** are descriptive characteristics of dimensions
    - Attribute hierarchies are top-down data organization used for aggregation and drill-down and rollup analyses

Star Schemas Performance Improvement

- Normalization of dimensional tables
- Maintaining multiple fact tables to represent different aggregation levels
- Denormalizing fact tables
- Partitioning and replicating fact tables

Online Analytical Processing (OLAP)

Databases geared towards BI that share three main characteristics

1. Multidimensional data analysis techniques
2. Advanced database support
3. User-interface made for non-technical users

SQL Extensions for OLAP

- Rollup
- Cube
- Materialized Views

Rollup

- Generates aggregates (sum, min, max, avg, etc.) by different dimensions
- Calculates the aggregate for each combination of dimensions, the sub-total for all columns except the last one listed and a grand total
- Usage: `GROUP BY ROLLUP(column1,...,columnn)`

Cube

- Also used to generate aggregates (sum, min, max, avg, etc.) by different dimensions
- Calculate the aggregate for each combination of dimensions, the sub-total for each column and a grand total
- Usage: `GROUP BY ROLLUP(column1,...,columnn)`

Materialized Views

- Unlike a regular view materialized views are storing data
- Materialized views are dynamic tables that are automatically updated when the underlying tables are updated
- Note that MariaDB doesn't support materialized views

Homework

- Read Chapter 16